

Online Temporal Action Detection for Distress Hand Signals in Continuous Video

Interim Progress Report

M.Tech Thesis — Panel Review

1. Introduction

1.1 Motivation

A person in danger is frequently unable to call for help by voice. They may be under surveillance by the person threatening them, in a public space where speech is not private, or physically prevented from speaking. A silent, deliberate hand gesture is in such cases the only channel available.

This observation is not hypothetical. The "Signal for Help" hand gesture, introduced in 2020 as a one-handed, speech-free indication of domestic distress, has since been used in real incidents and recognised by bystanders. Its effectiveness, however, depends entirely on a human being present, looking in the right direction, and knowing what the gesture means. Camera infrastructure is already widespread in the environments where such a signal would be made; what is absent is the ability to interpret it automatically.

This project addresses that gap: the automatic detection of distress hand signals in a continuous video stream, in real time.

1.2 Problem statement

The task is **online temporal action detection (OAD)**. Given a video stream arriving frame by frame, the system must assign a label to the **current** frame using only frames that have already arrived, and decide when to raise an alarm.

This is distinct from two superficially similar and much better-studied problems:

- **Action recognition** operates on pre-trimmed clips. The clip boundaries are given. In a deployed system no such boundaries exist, and establishing them is most of the difficulty.
- **Offline temporal action localisation** has access to the complete video and may refine a predicted boundary using later frames. A deployed alarm system cannot wait for the video to end.

The defining constraint of this work is therefore **causality**: no component may use information from a future frame. This constraint is easy to violate accidentally during development, because during development the entire video file is available. A model that peeks forward reports excellent accuracy and is useless in deployment.

1.3 Objectives

- 1 Design a strictly causal, two-stream (appearance and pose) online detector for hand gestures in continuous video.
- 2 Implement an explicit online decision layer governing when an alarm is raised, rather than treating a per-frame classifier output as an alarm.

- 3 Evaluate under a protocol appropriate to an alarm system — including false alarms per hour and onset-to-alarm latency, not accuracy alone.
- 4 Establish the result on a subject-disjoint split, with every reported number traceable to a configuration, a random seed, a frozen split file and a source-code revision.
- 5 Build the system so that the distress-signal dataset can be substituted later without modification to the model, training or evaluation code.

1.4 Scope and two-phase structure

Recording a purpose-built distress-signal dataset is a substantial undertaking and is deferred. The work is therefore structured in two phases.

Phase 1 (current) develops and validates the complete system using the **IPN Hand** dataset as an engineering vehicle. Its thirteen classes are touchless interface commands — pointing, clicking, swiping, zooming — and are explicitly *not* distress signals. It is used because it has the correct structural properties: continuous unsegmented video, a dominant non-gesture class, per-subject metadata permitting subject-disjoint evaluation, and gesture onsets that are genuinely ambiguous rather than cleanly annotated.

Phase 2 substitutes purpose-recorded distress-signal data behind a dataset adapter boundary, with no change required to downstream code.

This separation allows the methodological contributions to be developed and tested rigorously before the domain data exists.

1.5 Expected contributions

- 1 A strictly causal two-stream online detector, with causality enforced by automated test rather than asserted in prose.
- 2 An online decision layer treated as a first-class component and systematically ablated.
- 3 An evaluation protocol for alarm-oriented detection, reporting the latency / false-alarm trade-off as a curve rather than a single operating point.
- 4 A dataset-agnostic implementation permitting substitution of the target domain data.

2. Literature Review

2.1 Problem taxonomy

Video understanding tasks differ principally in what information is available at inference.

Task	Input at inference	Future frames	Representative work
Action recognition	Pre-trimmed clip	Available	Two-Stream, I3D, TSM
Offline temporal localisation	Whole video	Available	BSN, BMN
Online action detection	Stream, prefix only	Unavailable	TRN, OadTR, LSTR, MiniROAD
Action anticipation	Stream, predicts ahead	Unavailable	RED

Online action detection was formalised by De Geest et al. (2016), who introduced the task alongside the TVSeries dataset and demonstrated that methods performing well on trimmed clips degrade substantially

when boundaries are withheld.

2.2 Online action detection methods

Early approaches adapted recurrent architectures. **RED** (Gao et al., 2017) used a reinforced encoder–decoder for anticipation, establishing that modelling the near future improves present-frame classification. **TRN** (Xu et al., 2019) made this explicit by predicting future representations and feeding them back to inform the current decision — notably, without violating causality, since the predicted future is inferred rather than observed.

Subsequent work addressed the limited temporal horizon of recurrent models. **IDN** (Eun et al., 2020) learned to discriminate relevant from irrelevant accumulated information. **OadTR** (Wang et al., 2021) and **LSTR** (Xu et al., 2021) introduced transformer architectures, the latter separating long-term and short-term memory to extend effective context. **TeSTra** (Zhao and Krähenbühl, 2022) addressed the computational cost of attention over long histories using temporal smoothing kernels permitting streaming inference at constant cost per frame.

Most relevant to this work, **MiniROAD** (An et al., 2023) observed that much of the reported advantage of transformer approaches arises from a training/inference mismatch in recurrent baselines: during training, loss is typically applied at every position of a clip, whereas at inference only the final position is ever used. Applying non-uniform loss weighting — concentrating the objective at the final position — a minimal GRU attains competitive accuracy at a fraction of the computational cost. This finding directly motivates the architecture adopted here.

2.3 Online hand gesture detection

Work specific to continuous hand gesture streams is comparatively sparse. Molchanov et al. (2016) addressed online detection and classification of dynamic hand gestures with a recurrent 3D CNN and connectionist temporal classification, introducing the NVGesture dataset. Köpüklü et al. (2019) proposed a two-model architecture in which a lightweight detector gates a heavier classifier, explicitly addressing the single-time activation problem — ensuring one gesture produces exactly one output rather than a burst.

Benitez-Garcia et al. (2020) introduced **IPN Hand**, the dataset used in Phase 1 of this work, specifically to support continuous rather than trimmed evaluation. The dataset provides long sequences containing multiple gestures separated by labelled non-gesture intervals.

2.4 Feature representations

Online detectors conventionally operate on features extracted by a frozen, pretrained backbone rather than on raw pixels, which decouples representation learning from temporal modelling and makes experimentation tractable.

TSM (Lin et al., 2019) achieves temporal modelling at 2D CNN cost through channel shifting, and supports a causal (uni-directional) shift variant. **X3D** (Feichtenhofer, 2020) provides an efficient 3D architecture family. **VideoMAE** (Tong et al., 2022) and **VideoMAE V2** (Wang et al., 2023) apply masked autoencoding to video, yielding strong transferable representations; VideoMAE V2 is adopted here as the appearance backbone.

For the pose stream, **RTMPose** (Jiang et al., 2023) provides real-time whole-body keypoint estimation including hand keypoints, with **MediaPipe Hands** (Zhang et al., 2020) as a CPU-viable alternative.

2.5 Evaluation protocols

Per-frame mean average precision is the conventional OAD metric but is a poor proxy for alarm-system utility: it rewards correctly labelling the long interior of a gesture, which is of little operational interest once an alarm has been raised.

Point-level evaluation under the ODAS protocol instead scores whether a detection occurs within a tolerance window of the gesture onset, which corresponds more closely to the practical requirement. For an alarm system, two further quantities are decisive and are frequently omitted: the **false-alarm rate per unit time**, which determines whether the system is tolerable in deployment, and the **onset-to-alarm latency**, which determines whether it is useful. These trade off against one another, and reporting either in isolation is uninformative.

2.6 Research gap

Three gaps motivate this work.

- 1 **Automatic distress-signal detection is largely unaddressed.** The Signal for Help gesture is documented in public-safety literature, but computational work on detecting it in continuous video is minimal.
 - 2 **The decision layer is routinely neglected.** Published OAD work typically reports per-frame metrics and omits the mechanism converting a per-frame score sequence into discrete alarms — thresholding, persistence, refractory behaviour — despite this mechanism determining practical performance.
 - 3 **Alarm-oriented metrics are under-reported.** False alarms per hour and onset latency are seldom given, making published methods difficult to assess for deployment.
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3. Methodology

3.1 System overview

The pipeline comprises four stages:

- 1 **Feature extraction.** Each frame is represented by a feature vector from a frozen pretrained backbone, computed causally.
- 2 **Temporal model.** A causal recurrent network maintains state across the stream and emits per-frame class scores.
- 3 **Decision layer.** Per-frame scores are converted into discrete alarm events by causal smoothing, hysteresis thresholding, persistence and refractory logic.
- 4 **Evaluation.** Detected events are scored against ground truth under the protocol in §3.8.

3.2 Dataset and canonical format

Phase 1 uses IPN Hand: 200 videos, 50 subjects, 13 gesture classes plus a non-gesture class, 640×480 at 30 fps.

To permit later substitution of the distress-signal dataset, every dataset is converted by a small adapter into a single **canonical format** — per-frame feature arrays, an annotation table of closed frame intervals with subject identifiers, a class list, and a metadata file. All model, training and evaluation code reads only this format and never references a dataset by name. Adding a dataset requires one adapter and one configuration file.

A schema validator enforces the format. Its central invariant is that annotations must tile each video exactly: non-gesture intervals are stored as explicit labelled rows, never inferred from absence. This prevents the non-gesture class from silently disappearing, which would reduce the task to trimmed-clip classification.

3.3 The causality constraint

No component may use information from a future frame. This excludes bidirectional recurrent networks, centred convolutions and filters, unmasked full-sequence attention, and normalisation statistics computed over a complete sequence.

The constraint is enforced by an automated test rather than by inspection. The test supplies a truncated stream of length t and requires the outputs for frames $0 \dots t$ to be **bit-identical** to the corresponding slice of a full-stream run. Bit-identity rather than numerical closeness is required, since a module that is approximately causal is one that uses future information slightly, which is sufficient to bias a latency measurement.

To establish that the test itself has discriminative power, three deliberately non-causal implementations are constructed and the harness is required to **reject** each: a clip centred on its own index, a snippet grid anchored to the final frame of the video, and forward interpolation between snippets. All three produce output of correct shape, correct type and finite value — a test suite that has only ever been exercised against correct code establishes nothing.

3.4 Subject-disjoint splitting

All evaluation uses subject-disjoint splits: no subject appears in more than one of train, validation and test. Splitting by frame, window or clip would place frames of the same gesture instance on both sides of the split, and the resulting metric would measure memorisation of individuals rather than generalisation to new ones.

Splits are generated once, written to disk, and placed under version control. The test partition adopted is the one published by the dataset authors, which was verified to be subject-disjoint under the subject identity established in §4.3. Adopting rather than selecting the test partition removes the possibility that several candidate splits were evaluated before one was retained.

Cross-validation uses leave-one-subject-group-out (LOGO) over the non-test subjects. The primary validation set is defined as one of these folds, promoted, rather than drawn separately — otherwise those subjects would act as *training* subjects in the remaining folds and the variance estimate would characterise a different experiment from the headline result it is intended to qualify.

3.5 Feature extraction

Two parallel streams are extracted.

The **appearance stream** uses frozen VideoMAE V2 (Base). As the backbone consumes short clips rather than single frames, features are computed on a snippet grid and expanded to per-frame resolution. Each element of this scheme is dictated by causality: each clip spans frames *ending at* the frame it describes rather than centred upon it; the snippet grid is anchored at frame 0 rather than at the end of the video, so that the grid for a stream prefix is a prefix of the grid for the full stream; and values are **held forward** between grid positions rather than interpolated, since interpolation would incorporate a value computed from frames not yet arrived.

The **pose stream** extracts hand and upper-body keypoints, normalised per person by translation to a root joint and scaling by a reference length, so that the representation is invariant to subject distance from the camera.

Feature normalisation is deliberately *not* applied during extraction. Statistics fitted over a complete dataset constitute a form of leakage; mean and standard deviation are therefore fitted on the training split alone, at the stage that trains.

3.6 Proposed model

The proposed model (M1) is a causal gated recurrent network applied to the fused feature streams, following the MiniROAD finding that **non-uniform loss weighting** — applying the training objective only at the final position of each clip — aligns training with streaming inference and recovers most of the advantage

otherwise attributed to transformer architectures.

Further design elements:

- **Two-stream late fusion.** Separate heads on pose and appearance features with learned fusion of logits, with the fusion mechanism kept substitutable for ablation.
- **Temporal label smoothing.** Gesture onsets are ambiguous over approximately 0.3 s; targets are therefore ramped across the boundary rather than stepped.
- **Class-balanced objective.** The class distribution is severely skewed (§4.4).
- **Onset-aware clip sampling.** A fixed proportion of training clips is required to contain a gesture onset; without this constraint, onsets constitute a vanishing fraction of sampled positions and receive negligible gradient.

3.7 Online decision layer

Converting per-frame scores into alarms is treated as a distinct, ablatable component with all parameters exposed in configuration:

- **Causal smoothing** by exponential moving average or causal median. Centred filters are excluded, as they incorporate future frames.
- **Dual-threshold hysteresis** — an alarm is raised at a high threshold and sustained to a lower one, preventing oscillation near a single threshold.
- **k-of-n persistence** before firing, suppressing isolated spurious frames.
- **Refractory period** following an alarm, so that one gesture yields one alert.
- **Temperature calibration** on validation data, so that thresholds carry consistent meaning across models.

3.8 Evaluation protocol

The following are reported together for every configuration:

- per-frame mean average precision;
- point-level mean average precision (ODAS protocol), swept over onset tolerance from 1 to 10 seconds;
- event-level precision, recall and F1;
- **false alarms per hour;**
- **median and 95th-percentile onset-to-alarm latency;**
- end-to-end throughput and peak memory.

The headline figure plots latency against false alarms per hour, one curve per persistence setting, since no single operating point characterises an alarm system.

3.9 Reproducibility

Every run writes a directory recording the fully resolved configuration, the source-code revision and working-tree cleanliness, the random seeds, the software environment and the resulting metrics. Experiment tracking mirrors scalars to this directory, so that results remain traceable independently of any hosted service. No quantity affecting a result is hard-coded outside configuration files.

A working safeguard is applied throughout: any metric exceeding 97% is treated as evidence of leakage and investigated before it is reported. On a subject-disjoint online detection task such a value is implausible, and the characteristic failure mode of this class of project is not a poor result but an excellent and meaningless one.

4. Results and Evaluation

4.1 Status and scope of this section

Work to date has established the data foundations, the splitting methodology and the causality infrastructure. **Model training has not yet commenced, and no detection performance figures are available.** This section reports what has been evaluated: the integrity of the dataset, the correctness of the split, and the validity of the causality guarantee.

This ordering is deliberate. Each of the defects reported in §4.3 would have produced a plausible but incorrect performance figure rather than a visible error.

4.2 Dataset verification

Every published figure for IPN Hand was recomputed from the distributed files and compared against the values reported by Benitez-Garcia et al. (2020).

Quantity	Published	Computed	Agreement
Videos	200	200	exact
Subjects	50	50	exact
Gesture classes	13	13	exact
Gesture instances	4,218	4,218	exact
Non-gesture instances	1,431	1,431	exact
Frames	~800,000	800,491	within tolerance

Table II of the source publication was additionally reproduced row by row — per-class instance count, mean span duration and standard deviation — for all fourteen classes. This verifies the class *mapping* and not merely the class *count*; see §4.3.

4.3 Data integrity findings

Three defects were identified in the dataset or in the conventional means of reading it.

(i) Subject identity is not the subject index. IPN Hand video identifiers decompose into camera, subject, hand and clip fields. The subject field is not a global identity: it restarts at 1 for each camera and repeats across cameras, yielding only 29 distinct values for 50 subjects. The pair (camera, subject) yields exactly 50 groups of exactly 4 videos, consistent with the published figures, and the authors' own partition is disjoint under it while eight bare subject indices straddle it. Splitting on the index alone would have placed the same individual on both sides of the split.

(ii) The distributed frame counts are incorrect for fourteen videos. The supplied metadata reports 800,505 frames; the image files number 800,491. The discrepancy is exactly one frame for exactly fourteen videos, and those fourteen are precisely those whose frame directory contains a stray operating-system metadata file. The count was evidently produced by enumerating directory entries rather than images. Video length is therefore taken from the annotations, which agree with the image count for all 200 videos.

(iii) **The class-code mapping required verification.** The dataset distributes class *codes* without stating which code denotes which gesture; names were matched positionally against the published class table. Because an incorrect mapping would leave every aggregate count consistent, all 91 pairwise label exchanges were simulated, confirming that 90 would be detected by the per-class statistics. The single undetectable pair, for which the published duration figures round identically, was resolved by direct inspection of video frames.

A fourth defect was identified in the implementation itself. On first execution the causality suite rejected the feature extractor. The cause was not a future read: floating point matrix multiplication is not invariant to batch size, and a truncated stream produces a smaller final batch. The available responses were to relax the bit-identity requirement to a numerical tolerance, or to characterise and eliminate the dependency. The former would have left a causality test incapable of detecting a causality violation; the latter was adopted.

4.4 Dataset composition

The converted dataset comprises 800,491 frames across 200 videos and 50 subjects.

Class	Instances	Frames	% of frames
none	1,431	210,542	26.3
pointing with one finger	1,010	221,193	27.6
pointing with two fingers	1,007	225,683	28.2
click with one finger	200	11,197	1.4
click with two fingers	200	11,935	1.5
throw up	200	12,400	1.5
throw down	201	13,095	1.6
throw left	200	13,226	1.7
throw right	200	12,761	1.6
open twice	200	15,126	1.9
double click with one finger	200	13,525	1.7
double click with two fingers	200	13,910	1.7
zoom in	200	13,033	1.6
zoom out	200	12,865	1.6

The distribution has a consequence for method design. The non-gesture class is the largest by instance count but only the third largest by frame count: the two pointing classes are long resting or transitional states with a median duration of approximately seven seconds, rather than brief commands. Together these three classes account for roughly 82% of all frames, leaving the eleven command gestures approximately 18%. This bears directly on class balancing and on the false-alarm denominator, and the distress-signal data of Phase 2 is not expected to share this structure.

4.5 Split composition

Split	Subjects	Videos
Train	30	120
Validation	7	28
Test	13	52

Cross-validation uses five LOSGO folds of 7–8 subjects over the 37 non-test subjects. The split is frozen to disk, carries a digest of the subject list from which it was derived, and is refused at load time should that digest cease to match the dataset.

4.6 Verification status

Suite	Tests	Purpose
Causality	41	Hard constraint of §3.3, including three negative controls
Splits	36	Subject disjointness and fold integrity
Full suite	211	All of the above plus format, adapter and cache validation

All tests pass at the revision reported on the title page.

5. Remaining Work

Stage	Content	Status
0	Canonical data format and configuration system	Complete
1	Dataset acquisition, adapter, verification	Complete
2	Frozen subject-disjoint splits and LOSGO folds	Complete
3	Two-stream feature extraction and caching	In progress
4	Baselines and full evaluation harness	Not started
5	Proposed model (M1)	Not started
6	Online decision layer	Not started
7	Metrics and headline figure	Not started
8	Ablations and failure analysis	Not started

Stage 4 includes a sliding-window classifier baseline, implemented to a standard permitting fair comparison rather than as a strawman, since it represents the natural first approach to the problem and belongs in the ablation table.

Principal risk. Hardware for model training is not yet settled. Feature extraction is performed on a free cloud GPU service and the resulting feature cache is portable by design and verified by checksum; the hardware for Stages 5 to 8 remains the primary scheduling uncertainty.

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